

1. Suppose a particle's position is given by the equation $s(t) = 1 - t^2 + 5t$, where s is measured in meters and t is measured in seconds.
 - a) Find the average velocity of the particle during its first 4 seconds of motion. State answer with units.
 - b) Find the instantaneous velocity of the particle after 1 seconds. State answer with units.

2. Find $f'(x)$ if $f(x) = \frac{1-x}{2+x}$.

3. Find an equation that will tell you slope of any tangent line on the graph of $f(x) = 4x^2 - 9$.

4. Use the information found in problem #3 to construct the equation of a line that is tangent to the function $f(x) = 4x^2 - 9$ at the point where $x = \frac{1}{4}$.

5. Find $f'(x)$ if $f(x) = x^3 - 5x$.

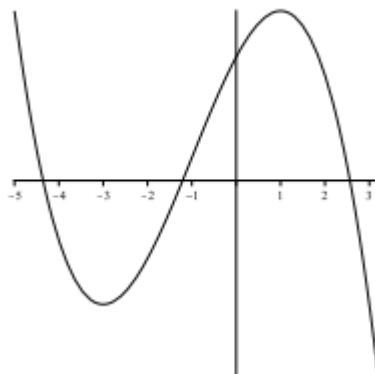
6. Find the derivative of $g(x) = \frac{1}{x+1}$.

7. Suppose that we know that $g(x)$ is an *even* function that is differentiable at $x=1$. we know that $g(x)$ has the following values:

$$g(-4)=1 \quad g(-2)=1 \quad g(-1)=3 \quad g(0)=3 \quad g(2)=1 \quad g(3)=1$$

Find $\lim_{x \rightarrow 1} g(x)$.

8. The graph of the function $f(x)$ is shown below.



One of the six graphs below is the graph of $f'(x)$. Circle the graph of $f'(x)$.

